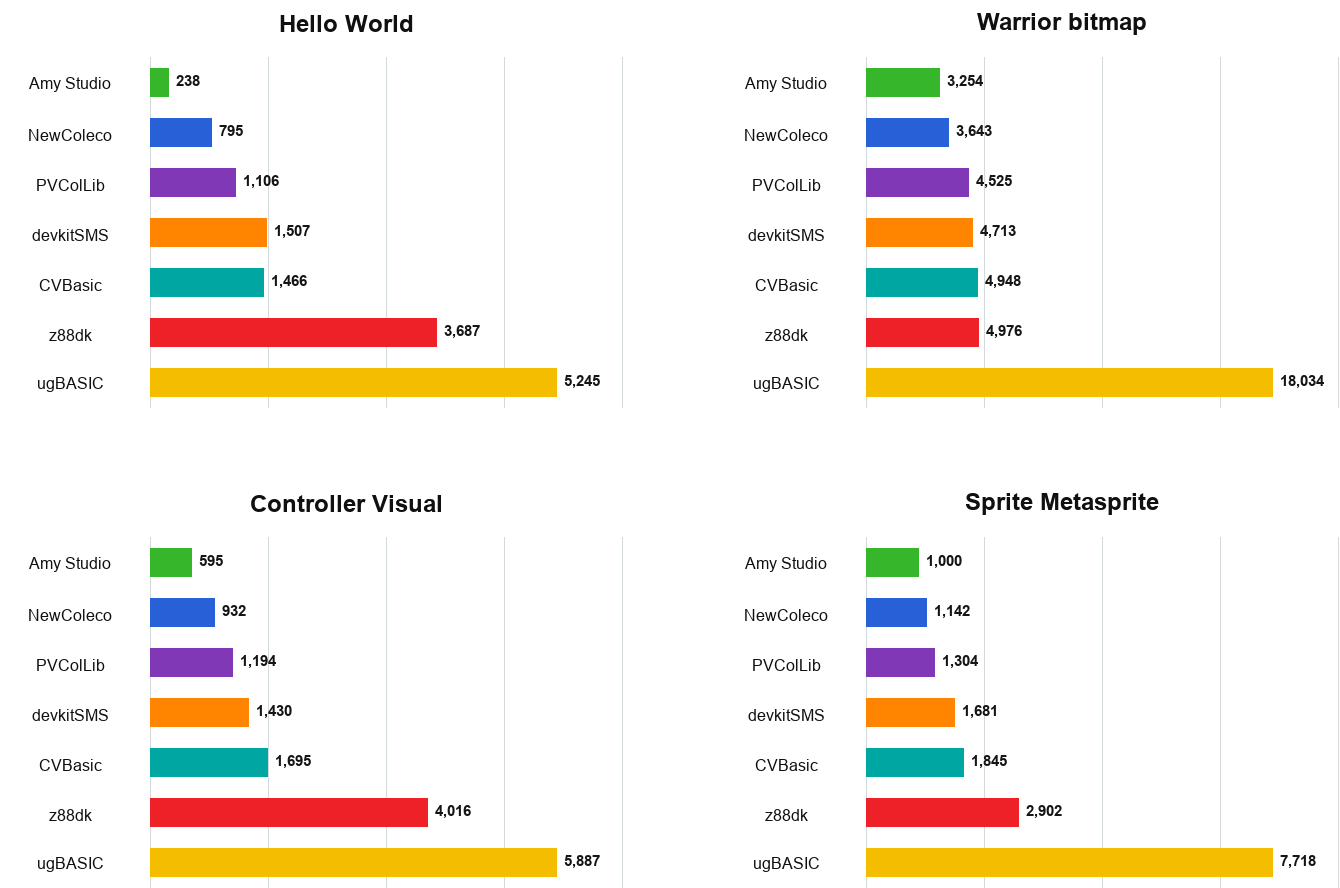


# Amy Studio ColecoVision Toolchain Comparison

Measured results, capabilities, compression evidence, and development priorities

## Real occupied size, excluding cartridge padding

sizes in bytes (lower is better); columns follow the four-sample total ranking



Occupied ROM size excludes cartridge padding. Lower is smaller, not automatically faster or better.

## Amy Studio source and GearColeco results

The compact listings are complete. The metasprite listing omits only its six pixel-pattern blocks, which remain in the evidence archive. Screenshots come from GearColeco 1.6.8 running each measured ROM for 180 NTSC frames.

### Hello World

```
text screen  
backdrop sky blue  
print centered at 11, "HELLO, WORLD!"  
screen on
```



GearColeco screenshot

GearColeco output after 180 NTSC frames

## Warrior Graphics II Bitmap

```
picture WarriorPicture:  
  pattern from "@project/warrior.pattern.zx0" codec zx0  
  color from "@project/warrior.color.zx0" codec zx0  
end picture  
  
show picture WarriorPicture
```



GearColeco screenshot

Exact GearColeco output; all seven tools reproduce the same target image

## Visual Controller Input

```
text screen
backdrop dark blue
print centered at 8, "CONTROLLER TEST"
print centered at 11, "MOVE OR PRESS A BUTTON"
print centered at 14, "KEYPAD IS PURPLE"
screen on
```

```
InputLoop:
  wait
  if keypad(1) <> 255 then
    backdrop magenta
  elseif joypad(1).fire then
    backdrop light red
  elseif joypad(1).up then
    backdrop light green
  elseif joypad(1).down then
    backdrop dark green
  elseif joypad(1).left then
    backdrop light yellow
  elseif joypad(1).right then
    backdrop cyan
  else
    backdrop dark blue
  end if
  goto InputLoop
```



GearColeco screenshot

GearColeco output at the neutral controller state

## Animated Three-Color Metasprite

```

u8 PlayerX = 120
u8 PlayerY = 88
u8 AnimationTick = 0
u8 AnimationFrame = 0

data PlayerMeta metasprite layers 3
  frame 0,15, 4,11, 8,1
  frame 12,15, 16,11, 20,1
end data

' Six 16x16 pattern blocks are included in the evidence archive.
sprites simple
set sprite count 3

MainLoop:
  wait
  if joypad(1).left and PlayerX > 1 then PlayerX -= 1
  if joypad(1).right and PlayerX < 239 then PlayerX += 1
  if joypad(1).up and PlayerY > 1 then PlayerY -= 1
  if joypad(1).down and PlayerY < 175 then PlayerY += 1
  AnimationTick += 1
  if AnimationTick = 8 then
    AnimationTick = 0
    AnimationFrame = 1 - AnimationFrame
  end if
  set metasprite PlayerMeta frame AnimationFrame to PlayerY,PlayerX using sprite 0
  update sprites
  goto MainLoop

```



GearColeco screenshot

A metasprite is one visual actor assembled here from three overlapping hardware sprites

## Graphics II compression corpus

All 42 unique 12,288-byte pictures passed exact round-trip checks with every codec. The gallery shows the actual TMS9918 output; the percentage is each picture's smallest compressed payload.

| Rank | Codec         | Total bytes | Median | Wins |
|------|---------------|-------------|--------|------|
| 1    | ZX0           | 166,556     | 23.63% | 22   |
| 2    | DAN2          | 166,923     | 23.99% | 5    |
| 3    | DAN3          | 167,186     | 23.99% | 9    |
| 4    | DAN1          | 167,240     | 24.04% | 6    |
| 5    | MegaLZ        | 171,016     | 24.66% | 0    |
| 6    | aPLib Compact | 171,484     | 24.52% | 0    |
| 7    | Pletter       | 171,751     | 24.80% | 0    |
| 8    | ZX7           | 171,788     | 24.77% | 0    |
| 9    | BitBuster     | 172,941     | 24.94% | 0    |
| 10   | ZX1           | 176,562     | 24.98% | 0    |
| 11   | ZX2           | 177,582     | 25.40% | 0    |
| 12   | LZF           | 193,412     | 28.36% | 0    |
| 13   | Nibble        | 214,555     | 35.48% | 0    |
| 14   | MDK-RLE       | 253,859     | 42.52% | 0    |

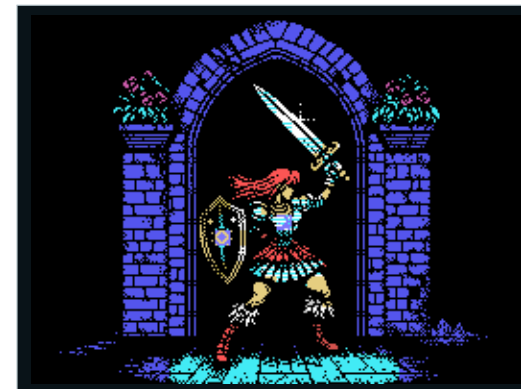
## Corpus gallery 1-6



**Cake**  
ZX0 - 24.07%



**Commando**  
ZX0 - 43.21%



**Warrior**  
ZX0 - 23.19%



**Barbarian**  
ZX0 - 34.42%

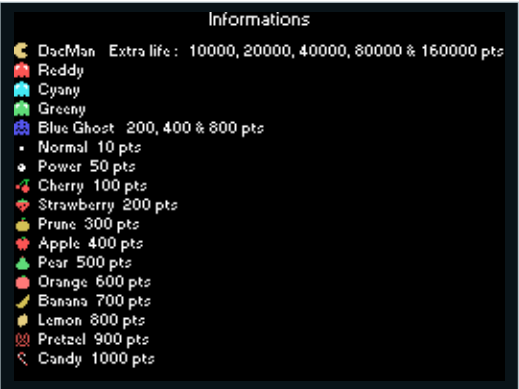


**421 title**  
ZX0 - 8.59%



**Dacman title**  
DAN1 - 7.89%

Corpus gallery 7-12



Dacman info  
ZX0 - 10.25%



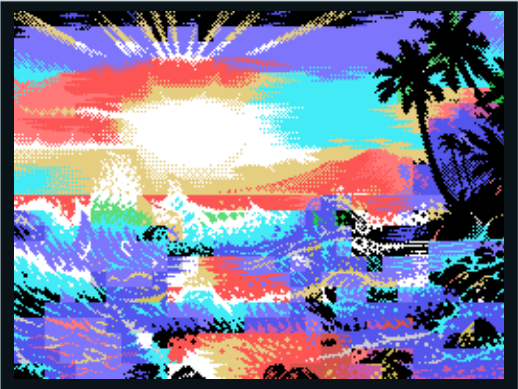
Dacman 2 title  
DAN2 - 9.51%



Chateau title  
ZX0 - 13.31%



Almighty God - Aliquis Acuario Ingens  
ZX0 - 58.87%



Aloha (ArtField 2010, 1)  
DAN1 - 56.88%



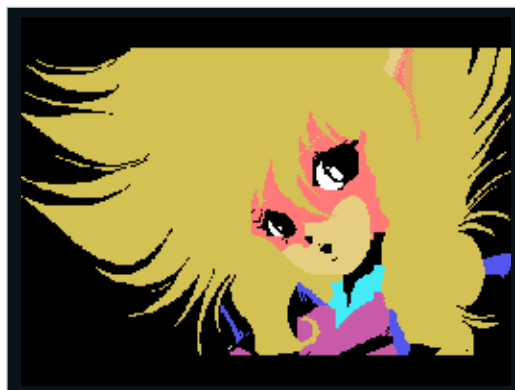
Aorante - Lady Saturn  
DAN3 - 29.73%



## Corpus gallery 13-18



bejeweled title public.pp  
ZX0 - 12.73%



benchmark-anime-portrait  
DAN2 - 17.55%



benchmark-band-logo  
ZX0 - 17.68%



benchmark-bat-character  
ZX0 - 7.98%



benchmark-blue-portrait  
ZX0 - 49.42%



benchmark-fantasy-camp  
DAN2 - 49.95%

## Corpus gallery 19-24



benchmark-feliz-navidad  
DAN3 - 33.76%



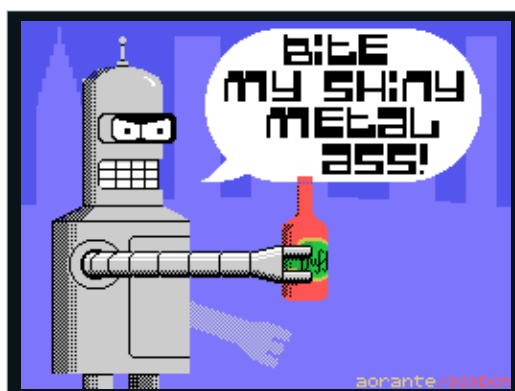
benchmark-laser-squad  
DAN2 - 60.94%



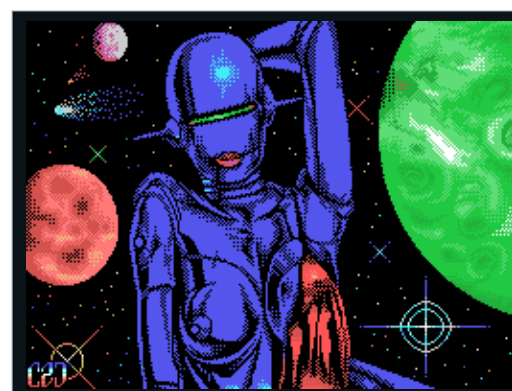
benchmark-mermaid-portrait  
ZX0 - 49.24%



benchmark-reservoir-dogs  
ZX0 - 18.84%



benchmark-robot-quote  
DAN3 - 16.89%

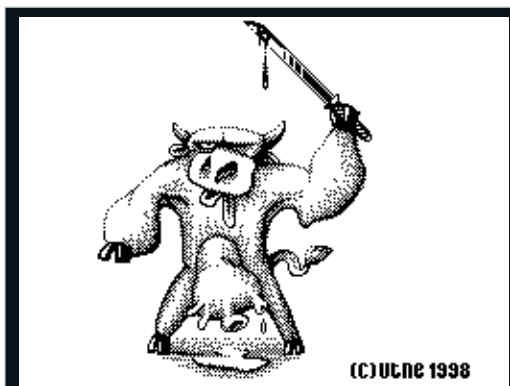


benchmark-space-soldier  
DAN2 - 45.97%

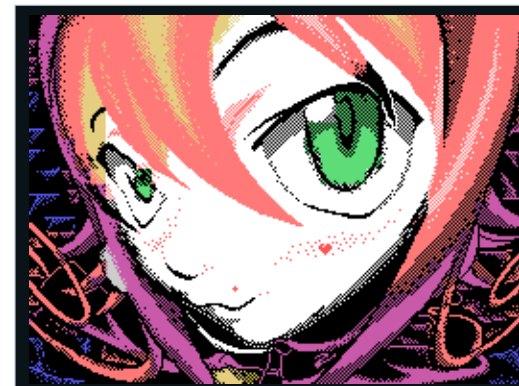
## Corpus gallery 25-30



benchmark-turban-character  
DAN1 - 64.36%



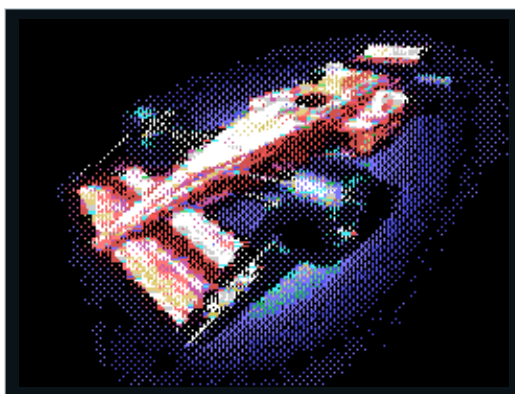
benchmark-wizard-sketch  
DAN3 - 17.67%



Blair (Chaos Constructions 2009, 6)  
ZX0 - 43.72%



Bugs (Demobit 1995, 4)  
DAN3 - 17.02%

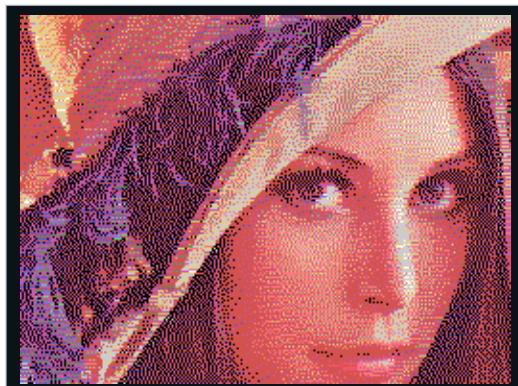


f1spp  
DAN3 - 34.81%



Kozmonaut (Forever 7, 7)  
DAN3 - 22.62%

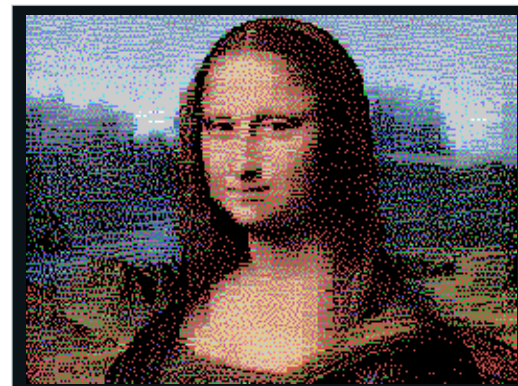
## Corpus gallery 31-36



lenna  
ZX0 - 76.50%



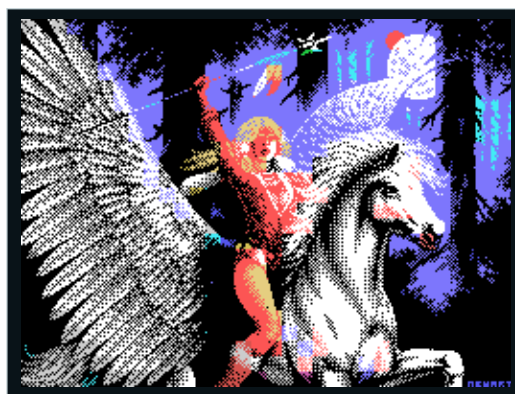
maze maniac.pp  
DAN3 - 22.05%



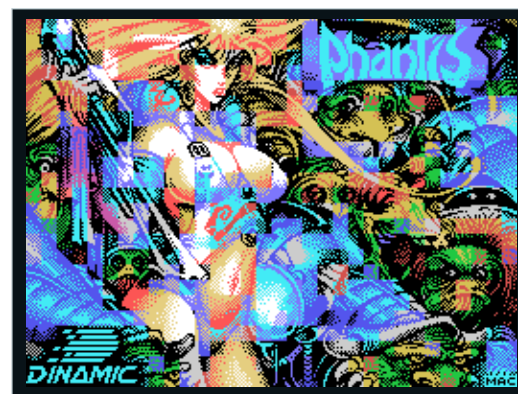
mona lisa  
ZX0 - 76.71%



NewColeco ROM File Edition  
ZX0 - 7.33%



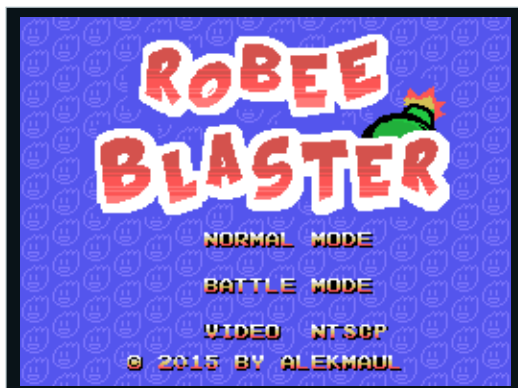
Pegasus (Millennium 1903, 1)  
ZX0 - 52.12%



Phantix (3BM OpenAir 2014, 1)  
DAN1 - 68.25%



## Corpus gallery 37-42



Robee Blaster title  
ZX0 - 16.94%



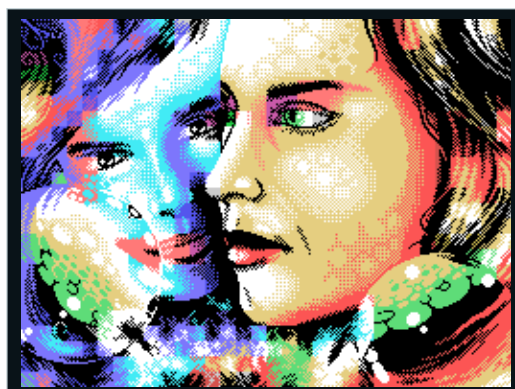
smurf challenge  
DAN1 - 10.29%



Stainly Rainbow (ArtField 2006, 3)  
ZX0 - 18.29%



Vranov (Chaos Constructions 2001, 7)  
ZX0 - 28.28%



Xzema (Chaos Constructions 2001, 2)  
DAN1 - 55.32%



Znachok (Enlight 1996, 15)  
DAN3 - 17.37%

Date: 2026-09-01

## Method

This comparison separates four kinds of evidence:

- **Runtime verified:** compiled for ColecoVision and checked in an emulator or self-test.
- **Build verified:** the ColecoVision source, header, library, or generated assembly was inspected.
- **Documented:** claimed by the tool's official documentation, but not yet measured here.
- **Unverified:** available somewhere in the wider tool, but not proven for ColecoVision.

Amy Studio combines a language and IDE; the others are mainly compilers or C kits. IDE-only features are listed separately from language capabilities.

## The seven solutions

| Solution             | Primary approach                                | ColecoVision position                                                            |
|----------------------|-------------------------------------------------|----------------------------------------------------------------------------------|
| Amy Studio           | ColecoVision-first Amy language and browser IDE | Original 1 KB RAM, BIOS-aware, no extra hardware by design                       |
| CVBasic              | Portable integer BASIC cross-compiler           | Mature native target; optional MegaCart and SGM support                          |
| z88dk +coleco        | General Z80 C toolchain                         | C compiler, assembler, linker, libraries, and app packaging                      |
| ugBASIC              | Portable retro BASIC compiler                   | Declared target with common high-level multimedia vocabulary                     |
| devkitSMS + sglib_cv | SDCC C plus compact game libraries              | ColecoVision adaptation of the SG/SMS development workflow                       |
| PVCoLib              | ColecoVision-focused SDCC C library and devkit  | Native VDP, controller, sprite, sound, music, and compression APIs               |
| NewColeco            | Historical SDCC C, CRTCV, CVLIB, and GETPUT 1.1 | Amy's pre-Studio ColecoVision workflow and direct ancestor of current techniques |

## Reproducible four-sample ROM suite

The suite builds four runnable programs with all seven solutions:

1. a visible Hello World;
2. the Warrior Graphics II bitmap picture;
3. a visual controller monitor;
4. an animated, controller-driven three-color metasprite.

A **metasprite** combines hardware sprites into one actor. This test overlaps three 16x16 layers (white, yellow, black), below the four-sprites-per-scanline limit.

The build scripts create 21 ROMs for the first three samples plus seven sprite ROMs, grouped under **build/competition.tools/report-five-tool-sample-sizes.ps1** records occupied sizes; the sprite script also runs the stricter GearColeco VRAM/SAT oracle.

Maximum native optimization used

| Solution                      | Setting used                                                          | Observation                                                                                         |
|-------------------------------|-----------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------|
| Amy Studio                    | Experimental                                                          | Saved 1 byte on Bitmap and 2 on Controller versus Balanced; Hello was unchanged                     |
| CVBasic 0.9.2                 | Normal compiler; TMSColor -z -p2 for Bitmap                           | No stronger compiler optimization switch was exposed; Pletter is used for its bitmap                |
| z88dk                         | +co1eco -03; ZX0 for Bitmap                                           | Coleco-safe direct-to-VRAM port of z88dk's ZX0 core; ZX7, MDKRLE, and raw baselines remain measured |
| ugBASIC 1.18                  | Default maximum of 16 peephole passes                                 | Explicit 32- and 64-pass builds produced the same Hello binary                                      |
| devkitSMS / SDCC 4.5          | --opt-code-size --max-allocs-per-node 100000 on the program and SGLib | Saved 60 bytes on Bitmap; Hello and Controller were unchanged                                       |
| PVColLib 1.6.0 / bundled SDCC | --opt-code-size --max-allocs-per-node 20000                           | Official build flags; linked only referenced library modules                                        |
| NewColeco / SDCC 3.8          | --std-c99; original prebuilt libraries plus historical DAN2           | The exact DAN2 bitmap is 626 bytes smaller than its GETPUT MDKRLE baseline                          |

These are the strongest **native settings validated here**. No competitor output passed through Amy's optimizer or MDL. Amy Experimental serves this size test; Balanced remains its default.

Real occupied size, excluding cartridge padding

| Sample            | Amy Studio | NewColeco | PVColLib | devkitSMS | CVBasic | z88dk  | ugBASIC |
|-------------------|------------|-----------|----------|-----------|---------|--------|---------|
| Hello World       | 238        | 795       | 1,106    | 1,507     | 1,466   | 3,687  | 5,245   |
| Warrior bitmap    | 3,254      | 3,643     | 4,525    | 4,713     | 4,948   | 4,976  | 18,034  |
| Controller Visual | 595        | 932       | 1,194    | 1,430     | 1,695   | 4,016  | 5,887   |
| Sprite Metasprite | 983        | 1,142     | 1,304    | 1,681     | 1,845   | 2,902  | 7,718   |
| Four-sample total | 5,070      | 6,512     | 8,129    | 9,331     | 9,954   | 15,581 | 36,884  |

Measured bitmap baselines: z88dk RAW 14,293 bytes, MDKRLE 5,911, ZX7 5,115, and ZX0 4,976; NewColeco GETPUT/MDKRLE 4,269 and DAN2 3,643. All reproduce both VRAM tables and 49,152 pixels. ugBASIC's **COMPRESSED** and **NONE** Warrior builds are identical at 18,034 bytes: MSC1 was discarded, and its RLE image branch is C128-only.

Legacy payloads are MDKRLE 3,687, DAN1 2,903, DAN2 2,897, and DAN3 2,891 bytes. DAN2 stays linked: DAN3 saves six payload bytes before decoder cost, while DAN2 is validated end to end.

**Occupied** includes header, runtime, code, and data, but excludes cartridge padding. Sources are: assembled length (Amy), **ROM\_END-\$8000** (CVBasic), unpadded binary (z88dk), generated code+data (ugBASIC), Intel HEX span above **\$8000** (devkitSMS/PVColLib), and complete linked binary (NewColeco).

Runtime and comparability verdict

| Sample            | Runtime result                                                  | Verdict                                 |
|-------------------|-----------------------------------------------------------------|-----------------------------------------|
| Hello World       | Seven ROMs complete 180 GearColeco frames                       | Stable startup                          |
| Warrior bitmap    | Seven native pipelines render the same 256x192 image            | 0 / 49,152 pixels differ                |
| Controller Visual | Six pass injected neutral, keypad, UP, FIRE, and release states | Partial: ugBASIC does not update VDP R7 |
| Sprite Metasprite | Seven pass the same VRAM and sprite-table checks                | Exact patterns, layers, and priority    |

PVColLib and NewColeco use \$F0 transparent-background text so VDP R7 changes remain visible; controller logic and occupied size are unchanged.

Native bitmap paths are Amy/z88dk ZX0, CVBasic Pletter, ugBASIC resources, devkitSMS aPLib, PVColLib RLE, and NewColeco DAN2. All framebuffers are exact despite equivalent internal table encodings. PVColLib Pletter failed VRAM validation and is excluded.

For z88dk, MDKRLE reaches VRAM at frame 93, ZX0 at 132, and ZX7 at 138. ZX0 saves 935 bytes versus MDKRLE and 139 versus ZX7; MDKRLE is faster. A transplanted SMS aPLib path failed and is excluded; native devkitSMS aPLib remains exact with frame interrupts disabled during decompression.

The initial ugBASIC mismatch came from the wrong RGB palette. With its target palette, all four corpus pictures render exactly.

Observations supported by this suite

- Amy is smallest for Hello, Controller, and this exact bitmap ROM (3,254 versus CVBasic's 4,948 bytes).
- Portable C/BASIC runtimes add visible fixed cost, but complete games may scale differently.
- Cartridge length includes packaging; a padded 32 KB file is not necessarily a 32 KB program.
- Smaller results count only after runtime validation; four samples cannot prove a universal winner.
- RAM, worst-frame cycles, build latency, sound, sprite pressure, and gameplay remain separate tests.



## Preliminary capability matrix

Legend: **Yes** = verified or explicit target support; **Partial** = manual or narrower support; **Pending** = must still be proven specifically on ColecoVision. \* marks a benchmark adaptation written here, not a native ColecoVision direct-to-VRAM path supplied by that solution.

| Capability                               | Amy Studio                                                                | CVBasic                         | z88dk                                                 | ugBASIC                                       | devkitSMS / SGLib_CV                         | PVCoLib                                     |
|------------------------------------------|---------------------------------------------------------------------------|---------------------------------|-------------------------------------------------------|-----------------------------------------------|----------------------------------------------|---------------------------------------------|
| 8/16-bit integers                        | Yes                                                                       | Yes                             | Yes                                                   | Documented                                    | Yes, through SDCC                            | Yes, through SDCC                           |
| 32-bit integers                          | Yes, runtime tested                                                       | No native BASIC type            | Yes                                                   | Documented; target cost pending               | Yes, through SDCC                            | Yes, through SDCC                           |
| Fixed-point                              | Yes, 8.8 and 16.16                                                        | No                              | Library/manual                                        | Documented; target proof pending              | Manual C/library code                        | Manual C/library code                       |
| Records/structures                       | Yes                                                                       | No                              | Yes                                                   | Documented custom types                       | Yes                                          | Yes                                         |
| 2D arrays                                | Yes                                                                       | No native 2D syntax             | Yes                                                   | Documented                                    | Yes                                          | Yes                                         |
| RAM overlays                             | Yes, first-class                                                          | Manual                          | C union/manual layout                                 | Pending                                       | C union/manual layout                        | C union/manual layout                       |
| Functions with values and returns        | Yes                                                                       | Limited BASIC procedures/DEF FN | Yes                                                   | Yes                                           | Yes                                          | Yes                                         |
| Inline Z80 assembly                      | Yes                                                                       | Yes                             | Yes                                                   | Documented                                    | Yes                                          | Yes                                         |
| TMS9918 screen and tile operations       | Yes                                                                       | Yes                             | Yes                                                   | Yes, target proof incomplete                  | Yes                                          | Yes                                         |
| Hardware sprite API                      | Yes                                                                       | Yes                             | Low-level/library                                     | Yes, exact target surface pending             | Yes                                          | Yes, 32-entry SAT                           |
| Four-sprite scanline mitigation          | Yes, stable ranges and flicker                                            | Yes, sprite flicker             | Manual                                                | Pending                                       | Manual/library-dependent                     | Yes, NMI sprite swapping                    |
| Keypad                                   | Yes                                                                       | Yes                             | Library/manual                                        | Pending                                       | Yes/target library                           | Yes, both ports                             |
| Spinner / Roller Controller              | Yes                                                                       | Yes                             | Not confirmed                                         | Pending                                       | Not confirmed                                | Yes, both ports                             |
| Held, pressed, and released input        | Yes                                                                       | Held; edges are manual          | Manual                                                | Manual edges over JOY                         | Yes, computed from NMI snapshots             | Manual edges over NMI snapshots             |
| Coleco PSG sound                         | BIOS tables, Tiny Sound, DSOUND                                           | Sound/music commands            | Sound libraries                                       | Sound commands, target proof pending          | PSGlib_CV                                    | BIOS-style sound tables and sequenced music |
| Direct-to-VRAM compression               | Fourteen active codecs, including MegaLZ; ZX0 Classic measured separately | Pletter                         | RAM APIs; ZX0, ZX1, ZX2, and ZX7 benchmark VRAM ports | Resource conversion; RAM-oriented compression | ZX7 and aPLib                                | RLE, Pletter, DAN1/2/3                      |
| ROM banking                              | Intentionally no                                                          | Yes                             | Yes                                                   | Pending                                       | SMS workflow has banking; CV support pending | MegaCart tools and examples                 |
| Source-level Coleco debugger             | Integrated                                                                | External emulator               | External debugger/emulator                            | External or IDE-dependent                     | External debugger/emulator                   | External debugger/emulator                  |
| Rewind, breakpoints, VRAM/RAM inspection | Integrated                                                                | External                        | External                                              | External                                      | External                                     | External                                    |
| Graphics editors and project assets      | Integrated                                                                | Separate tools                  | Separate tools                                        | Conversion-oriented IDE/tools                 | Separate asset tools                         | gfx2co1 and separate asset tools            |
| Automated ROM behavior tests             | Integrated project pipeline                                               | Not supplied as one system      | Can be assembled manually                             | Not established                               | Not established                              | Not established                             |

## Detailed language and runtime evidence

### Data model

| Area                    | Amy Studio                                                 | CVBasic                  | z88dk                               | ugBASIC                                                                  | devkitSMS                    | PVColLib                     |
|-------------------------|------------------------------------------------------------|--------------------------|-------------------------------------|--------------------------------------------------------------------------|------------------------------|------------------------------|
| Integer widths          | Explicit signed/unsigned 8/16/32                           | 8/16, signed or unsigned | C 8/16/32                           | Backend defines signed/unsigned 8/16/32                                  | SDCC C 8/16/32               | SDCC C 8/16/32               |
| Fractional math         | <b>fixed</b> , <b>ufixed</b> , <b>fixed32</b> , <b>fp5</b> | None                     | C/library float and fixed libraries | <b>VT_FLOAT</b> exists; Coleco cost pending                              | SDCC float/manual fixed      | SDCC float/manual fixed      |
| Decimal scores          | Packed BCD 1-12 digits                                     | Manual                   | Manual/library                      | Manual                                                                   | Manual/library               | Manual/library               |
| Aggregates              | Nested records, record arrays                              | None                     | struct/union                        | Typed arrays and internal resource types; user-structure surface pending | struct/union                 | struct/union                 |
| Multidimensional arrays | Primitive 2D arrays                                        | One-dimensional only     | Native C                            | Typed arrays; dimensional coverage pending                               | Native C                     | Native C                     |
| Mutually exclusive RAM  | First-class overlays/scenes                                | Manual aliases           | union/linker/manual                 | Banking/resource allocator; no equivalent proven                         | union/linker/manual          | union/linker/manual          |
| Dynamic strings         | Deliberately fixed-buffer oriented                         | Mostly literals/printing | C strings and allocation libraries  | First-class strings                                                      | C strings; risky in 1 KB RAM | C strings; risky in 1 KB RAM |

Amy's data model is the most game-oriented for stock ColecoVision RAM. C remains the most general, but generality does not provide automatic lifetime analysis or a debugger-aware overlay. ugBASIC has a broad internal type system; its actual ColecoVision ROM/RAM cost still requires compilation.

### Control, timing, and code organization

| Area                 | Amy Studio                                    | CVBasic                                                              | z88dk                      | ugBASIC                                                                 | devkitSMS                               | PVColLib                                     |
|----------------------|-----------------------------------------------|----------------------------------------------------------------------|----------------------------|-------------------------------------------------------------------------|-----------------------------------------|----------------------------------------------|
| Procedures/functions | Typed values, <b>ref</b> , returns, recursion | Procedures without value parameters; <b>DEF FN</b> expression macros | Full C                     | Procedures/functions documented                                         | Full C                                  | Full C                                       |
| State machines       | Typed zero-RAM dispatch                       | Manual <b>SELECT</b> /labels                                         | Manual enum/switch/table   | Manual control constructs                                               | Manual enum/switch/table                | Manual enum/switch/table                     |
| VBlank               | <b>wait</b> , frame hooks, timers             | <b>WAIT, ON FRAME</b>                                                | HALT/NMI hooks/manual      | <b>WAIT VBL, EVERY</b> , timers                                         | <b>SG_waitForVBlank</b> , frame handler | <b>vdp_waitvblank</b> , user <b>nmi</b> hook |
| Parallel animation   | Not yet first-class                           | Manual                                                               | Manual/interrupt           | <b>ANIMATION, ANIMATE, MOVE</b> , paths and multitasking source present | Manual frame service                    | Manual NMI/frame service                     |
| Inline assembly      | Amy ASM bridge                                | <b>ASM, CALL, USR</b>                                                | Native inline/external ASM | Supported                                                               | SDCC inline/external ASM                | SDCC inline/external ASM                     |

ugBASIC is the strongest source of ideas for a future optional Amy animation service. It also shows the risk: animations create state variables, thread handles, paths, delays, signals, and background-preservation storage. Amy should not copy that surface until exact RAM and cycle costs are visible and the entire service disappears when unused.

### TMS9918 graphics and sprites

| Area                   | Amy Studio                                 | CVBasic                        | z88dk                              | ugBASIC                                 | SGlib_CV                             | PVColLib                             |
|------------------------|--------------------------------------------|--------------------------------|------------------------------------|-----------------------------------------|--------------------------------------|--------------------------------------|
| Text/tile/bitmap modes | Coleco-specific commands and editors       | Modes and direct VRAM commands | Generic TMS9918/graphics libraries | TMS9918 backend and resource conversion | Tile, bitmap, VDP and VRAM functions | Text, bitmap, VDP and VRAM functions |
| Pixel/line/circle      | Yes, programmer controls NMI-safe batching | Yes                            | Generic graphics                   | TMS9918 drawing backend                 | Pixel/bitmap helpers                 | Low-level VRAM/bitmap support        |

| Area              | Amy Studio                                  | CVBasic              | z88dk             | ugBASIC                                  | SGlib_CV                             | PVColLib                                         |
|-------------------|---------------------------------------------|----------------------|-------------------|------------------------------------------|--------------------------------------|--------------------------------------------------|
| Sprites           | Fields, clipping helpers, movement, editor  | Define/place sprites | Low-level/library | Sprite conversion and TMS9918 operations | SAT builder, clipping, finalize/copy | 32-entry SAT, update and fast-update APIs        |
| Collision         | Hitboxes, tile groups, hardware status      | VDP status/manual    | Hardware/manual   | Hardware collision/hit backend           | VDP status/manual                    | <code>spr_collide</code> hitboxes and VDP status |
| Scanline overflow | Explicit flicker and stable priority ranges | Sprite flicker       | Manual            | No verified policy                       | Manual ordering                      | <code>spr_getentry</code> swaps sprites each NMI |
| Asset authoring   | Integrated bitmap/tile/sprite/frame editors | Separate tools       | Separate tools    | Automatic image conversion               | External asset tools                 | <code>gfx2col</code> and external tools          |

Amy's lead is the complete authoring/debugging workflow and explicit TMS9918 policy. ugBASIC's automatic modern-image conversion is substantial, while SGlib\_CV offers a compact C SAT workflow.

## Controllers

| Capability                | Amy Studio               | CVBasic               | z88dk                                | ugBASIC                                                 | SGlib_CV                     | PVColLib                                         |
|---------------------------|--------------------------|-----------------------|--------------------------------------|---------------------------------------------------------|------------------------------|--------------------------------------------------|
| Two directions/fire ports | Yes                      | Yes                   | <code>joystick(1/2)</code>           | <code>JOY(0/1)</code> backend                           | NMI snapshots both ports     | <code>joypad_1/2</code> , four fire bits         |
| Two keypad ports          | Yes                      | Yes                   | <code>joystick(3/4)</code> high byte | <code>SCANCODE</code> path; exact port behavior pending | <code>SG_readCVNumPad</code> | <code>keypad_1/2</code>                          |
| Pressed/released          | First-class per property | Manual previous state | Manual previous state                | Manual previous state                                   | First-class status helpers   | Manual previous-state comparison                 |
| Spinner/Roller            | Yes                      | Yes                   | Not found in Coleco target           | Not found yet                                           | Not found                    | <code>spinner_1/2</code> , reset and enable APIs |
| Automatic minimal backend | Yes, capability selected | Runtime-integrated    | Linker includes referenced routines  | Deployment system includes used modules                 | Library/NMI input service    | Linker extracts referenced modules               |

## Sound and music

| Area                | Amy Studio                                                                                               | CVBasic                            | z88dk                      | ugBASIC                    | devkitSMS                          | PVColLib                                     |
|---------------------|----------------------------------------------------------------------------------------------------------|------------------------------------|----------------------------|----------------------------|------------------------------------|----------------------------------------------|
| SN76489 tones/noise | BIOS sound tables and direct formats                                                                     | <b>SOUND</b>                       | PSG libraries/manual ports | SN76489 backend            | PSGlib_CV                          | BIOS-style sound tables                      |
| Sequenced music     | BIOS format and Tiny Sound                                                                               | <b>MUSIC</b> , simple/full players | External/player libraries  | <b>MUSIC</b> backend       | PSG streams; looping/status        | NMI-serviced music sequences                 |
| Concurrent SFX      | Coleco table areas/Tiny Sound                                                                            | Channel depends on music mode      | Library-dependent          | Pending measurement        | PSG SFX channels and frame service | Four music areas plus sound areas 5+ for SFX |
| Digital samples     | DSOUND                                                                                                   | No first-class support             | Manual/custom              | Pending                    | No first-class support found       | No first-class support found                 |
| Authoring           | Integrated table inspector, tone preview, echo-tail builder, MIDI capture; full sequence editing pending | Note-oriented BASIC source         | External VGM/tools         | Source/resource conversion | External PSG tools                 | External table/asset tools                   |

Amy has the broadest playback formats, while CVBasic keeps the simplest music source syntax. Amy Studio can now inspect existing BIOS tables, preview generated commands, audition steady/fade/echo envelopes, and capture note, velocity, and duration from Web MIDI. The remaining authoring gap is a complete multi-command sequence editor with byte-exact source/project write-back.

## Compression evidence

Compression must be scored as **compressed payload + linked decompressor**, with destination and cycles. Counting a host compressor without a ColecoVision decoder is invalid.

All fourteen integrated Amy codecs were also compiled as separate Balanced ROMs and run for 180 NTSC frames in GearColeco. Every ROM reproduced Warrior's 6,144 Pattern bytes and 6,144 Color bytes exactly in VRAM. Run `node tools/test-integrated-codec-vram-roms.mjs` to repeat this check.

## Direct-to-VRAM speed ranking

This benchmark measures two complete 6,144-byte Amy decompression commands (Pattern and Color) with NMI disabled. GearColeco counts exact Z80 master-clock cycles between executable markers and then verifies all 12,288 VRAM bytes. The samples are the mechanically selected most-compressible, median, and least-compressible corpus pictures. Run `node tools/benchmark-codec-vram-cycles.mjs`.

| Rank | Codec         | Decoder bytes | Average cycles | Observed range      | NTSC frames | PAL frames |
|------|---------------|---------------|----------------|---------------------|-------------|------------|
| 1    | MDK-RLE       | 46            | 445,166        | 397,701-480,470     | 7.45        | 6.24       |
| 2    | LZF           | 117           | 2,228,378      | 1,699,652-2,499,555 | 37.30       | 31.23      |
| 3    | Nibble        | 115           | 2,500,641      | 1,109,539-4,605,294 | 41.86       | 35.04      |
| 4    | BitBuster     | 166           | 2,534,837      | 2,200,323-2,848,558 | 42.43       | 35.52      |
| 5    | ZX2           | 115           | 2,640,553      | 2,523,782-2,813,703 | 44.20       | 37.00      |
| 6    | ZX1           | 127           | 2,649,233      | 2,542,465-2,824,788 | 44.35       | 37.12      |
| 7    | ZX0           | 133           | 2,776,094      | 2,593,812-2,917,877 | 46.47       | 38.90      |
| 8    | aPLib Compact | 244           | 2,882,985      | 2,038,419-3,910,905 | 48.26       | 40.40      |
| 9    | Pletter       | 212           | 3,011,986      | 2,593,248-3,420,132 | 50.42       | 42.21      |
| 10   | ZX7           | 136           | 3,045,957      | 2,646,525-3,421,800 | 50.99       | 42.68      |
| 11   | DAN3          | 205           | 3,260,654      | 2,819,207-3,514,505 | 54.58       | 45.69      |
| 12   | MegaLZ        | 162           | 3,545,056      | 3,003,617-4,081,177 | 59.35       | 49.68      |
| 13   | DAN2          | 212           | 3,683,858      | 2,806,792-4,725,242 | 61.67       | 51.62      |
| 14   | DAN1          | 205           | 3,809,391      | 2,827,513-5,013,133 | 63.77       | 53.38      |

Frame values are time equivalents (**59, 736** cycles NTSC; **71, 364** PAL), not VBlank waits. Speed and ROM-size rankings are separate: choose using compressed payload plus decoder size, then check whether the measured loading time suits the game. Nibble and aPLib vary substantially by input, which is why Amy Studio should present per-asset estimates rather than one universal winner.

| Solution   | Confirmed formats                                                                     | Direct VRAM status                                                           | Integrated selection                                  |
|------------|---------------------------------------------------------------------------------------|------------------------------------------------------------------------------|-------------------------------------------------------|
| Amy Studio | ZX0, ZX1, ZX2, aPLib, MegaLZ, ZX7, Pletter, DAN1/2/3, LZF, BitBuster, MDK-RLE, Nibble | ColecoVision paths, including workspace-based formats                        | Browser comparison/import and asset metadata          |
| CVBasic    | Pletter                                                                               | <b>DEFINE CHAR/COLOR/SPRITE/VRAM PLETTER</b>                                 | Explicit source keyword                               |
| z88dk      | ZX0/1/2/7 and aPLib families, multiple speed/size decoders                            | Stock decoders target RAM; ZX0 and ZX7 Coleco VRAM adaptations verified here | Manual headers/linking and host tools                 |
| ugBASIC    | MSC1 and RLE types in compiler source                                                 | MSC1 image fallback verified; RLE is not implemented for Coleco              | Resource compiler can choose compression when it wins |

| Solution                  | Confirmed formats      | Direct VRAM status                                                               | Integrated selection                              |
|---------------------------|------------------------|----------------------------------------------------------------------------------|---------------------------------------------------|
| SGlib_CV / SMSlib routine | ZX7 and aPLib          | ZX7 API; aPLib direct-to-VRAM exact with frame IRQ disabled                      | Manual host compression and C asset inclusion     |
| PVColLib                  | RLE, Pletter, DAN1/2/3 | RLE direct-to-VRAM verified exactly here; other paths remain separate candidates | <b>gfx2col</b> conversion and C asset inclusion   |
| NewColeco                 | GETPUT 1.1 MDK-RLE     | Direct-to-VRAM verified exactly on Warrior                                       | External CVPaint/asset tools and C data inclusion |

z88dk's public ZX0 integration request dates from February 2021, consistent with ZX0/ZX7 being available there by spring 2021. Its bundled compressor identifies itself as ZX0 v1.5 and emits the official classic v1 format. Amy Studio's current **zx0** codec emits the later official v2 format by default. The streams are intentionally not interchangeable. z88dk's stock routines target RAM; this benchmark adds only ColecoVision TMS9918 direct-to-VRAM adaptations and validates their output independently.

## Codec naming and integration policy

| Candidate   | Honest Amy name                                       | Current evidence                                                                              | Integration condition                                     |
|-------------|-------------------------------------------------------|-----------------------------------------------------------------------------------------------|-----------------------------------------------------------|
| ZX0         | <b>ZX0</b> / codec <b>zx0</b>                         | Existing v2 browser encoder and Coleco VRAM decoder                                           | Keep as the default                                       |
| ZX0 classic | <b>ZX0 Classic (v1)</b> / proposed codec <b>zx0v1</b> | z88dk v1.5 compressor plus exact benchmark VRAM port                                          | Explicit extension and cross-format rejection tests       |
| ZX1         | <b>ZX1</b> / codec <b>zx1</b>                         | Byte-identical browser encoder; exact GearColeco VRAM and cycle proof                         | Integrated and measured                                   |
| ZX2         | <b>ZX2</b> / codec <b>zx2</b>                         | Exact corpus round-trip, five-profile VRAM proof, and cycle proof                             | Integrated and measured                                   |
| aPLib       | <b>aPLib</b> / codec <b>aplib</b>                     | Bidirectional appack parity and exact Amy/GearColeco VRAM ROM                                 | Integrated; keep NMI-safe upload and attribution explicit |
| MegaLZ      | <b>MegaLZ</b> / codec <b>megalz</b>                   | Independent DEC40-compatible encoder; exact corpus round-trip and GearColeco VRAM/cycle proof | Integrated and measured                                   |
| MSC1        | <b>MSC1</b>                                           | ugBASIC discards it for Warrior when it gives no gain                                         | Useful Coleco corpus wins and a VRAM strategy             |

A host compressor alone is insufficient. An Amy codec requires round-trip tests, exact GearColeco VRAM output, decoder cost, explicit destination semantics, malformed-stream failure tests, documentation, and attribution. RAM-only APIs and locally written VRAM ports must remain labelled.

Current verdict: Amy leads in codec breadth and ColecoVision workflow; devkitSMS has verified ZX7 and aPLib paths, PVColLib has a verified RLE path, and CVBasic has a concise Pletter path. On Warrior and Cake, official ZX1, ZX2, and aPLib do not beat Amy's ZX0 first-use ROM size. aPLib nevertheless cuts the devkitSMS Warrior ROM from 13,680 raw bytes to 4,713 occupied bytes. ZX1 is now an integrated Amy codec with a 127-byte direct-to-VRAM decoder and exact runtime and cycle proofs. ugBASIC's MSC1 is a valid comparison candidate, not ten separate codecs, but it does not improve the Warrior resource.

## What MSC1 actually is

MSC1 is ugBASIC's compact block-repetition format, not another general ZX0-style LZ codec. A stream contains literal blocks of 1-127 bytes and back-references that repeat one four-byte sequence up to 32 times. The back-reference offset is limited to roughly 1 KB, and a zero token ends the stream. This favors maps, records, planar graphics, and other data with repeated groups of four bytes.

The current Z80 decoder is simple and Apache-2.0 licensed, but writes to ordinary memory with **LD (DE), A**. On a stock 1 KB ColecoVision it cannot directly expand a 6 KB pattern or color table. More importantly, ugBASIC's generic **LOAD("file") COMPRESSED** implementation only invokes MSC1 when the target declares expansion banks. The standard Coleco target does not, so the keyword is accepted but ignored there. Five raw/compressed resource pairs produced byte-identical generated code and data. This is therefore not presently a general Coleco capability comparable to Amy's direct-to-VRAM codecs.

MSC1 can still be selected by some ugBASIC image/resource conversion paths. A fair evaluation of that narrower feature needs a direct-to-VRAM proof and a mixed corpus: bitmap tables, name tables, tile sets, sprite frames, level maps, record arrays, and sound data. Only candidates that reduce **payload + linked decoder**, round-trip exactly, and pass GearColeco should be added.

## aPLib integration status

aPLib has three useful pieces already available locally: the official host compressor, Amy's JavaScript beam-search encoder, and direct-to-VRAM Z80 implementations in z88dk/devkitSMS. The devkitSMS routine uses the Coleco-compatible **\$BF/\$BE** VDP ports, but is explicitly unsafe while the display is active and lacks the Coleco BIOS protection used by **SG\_decompressZX7toVRAM**.

The JavaScript encoder now passes bidirectional differential checks against official aPLib on real TMS9918 tables and structured, incompressible, repetitive, short, and boundary-sized data. Its exact-cost planner produces 3,054 bytes for Cake and 2,973 for Warrior, versus official raw payloads of 3,088 and 2,975. Official **appack** still wins some individual resources, so neither encoder universally dominates.

Amy now ships the browser encoder and a 244-byte compact public-domain SMSlib-derived ColecoVision VRAM decoder. **decompress aplib Source to vram.pattern** accepts a raw **.aplib** stream and the existing VRAM upload wrapper provides the same NMI-safe contract as other Amy LZ codecs. A compiled Amy ROM reconstructed Warrior's 6,144-byte pattern and color tables exactly in GearColeco under all five optimization profiles. The Balanced test ROM occupied 3,517 bytes with 2,973 compressed data bytes. The official aPLib executable is used only for differential testing and is not redistributed.

The first reproducible size results and the rules for the cross-tool runtime comparison are in **competition/benchmarks/compression/README.md**. On two real 12,288-byte TMS9918 pictures, ZX0 has the smallest Amy first-use ROM total after its current 133-byte decompressor estimate is included. The separate GearColeco cycle benchmark supplies the speed evidence; size and speed remain independent selection criteria.

Memory, ROM size, and banking

| Area                    | Amy Studio                           | CVBasic                     | z88dk                      | ugBASIC                                    | devkitSMS                                | PVColLib                            |
|-------------------------|--------------------------------------|-----------------------------|----------------------------|--------------------------------------------|------------------------------------------|-------------------------------------|
| Stock 1 KB RAM focus    | Core design, RAM estimates, overlays | Yes, global/static model    | Configurable CRT/C runtime | Backend manages runtime/resources          | SDCC/static library model                | SDCC/static library model           |
| Dead helper elimination | Capability-driven generation         | Compiler-generated runtime  | Linker sections/libraries  | Deploy-on-use modules and target optimizer | Linker library extraction                | Linker library extraction           |
| Optimizer               | Five profiles plus runtime corpus    | Z80 optimizer and peepholes | sccz80/zsdcc optimizers    | Coleco-specific optimizer source           | SDCC optimizer/peepholes                 | SDCC size optimization              |
| Beyond 32 KB            | Deliberately excluded                | MegaCart up to 1 MB         | Coleco banking/toolchain   | Target support not yet proven              | MegaCart and banked functions documented | MegaCart tools and example verified |
| Debug-aware RAM names   | Yes, including overlay aliases       | Assembly labels             | Map/debug symbols          | Generated symbols                          | Map symbols                              | Map symbols                         |

Banking is a capability advantage for CVBasic, z88dk, and devkitSMS, but not a feature Amy intends to adopt: Amy explicitly preserves the original unexpanded-hardware philosophy. The comparison should describe this limitation honestly without turning it into a roadmap requirement.

Stock baseline versus expanded hardware

The size and runtime ranking above targets an original ColecoVision: TMS9918A video, SN76489 sound, 1 KB system RAM, Coleco BIOS, and an unbanked cartridge whenever possible. Optional hardware is credited separately because it changes the machine available to the programmer:

| Extension                        | Verified or documented advantage                                                        | Scope in this comparison                                                                       |
|----------------------------------|-----------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------|
| MegaCart / bank switching        | CVBasic, z88dk, devkitSMS, and PVColLib can exceed the normal cartridge space           | Valid capability; excluded from stock-ROM size ranking                                         |
| F18A                             | PVColLib provides dedicated APIs and examples for enhanced video hardware               | Valid for modified or compatible clone systems; out of scope for ColecoVision-exclusive titles |
| SGM / AY-3-8910-compatible sound | CVBasic and PVColLib provide explicit SGM paths; PVColLib also detects SGM RAM and ADAM | Valid expansion/clone capability; out of scope for the stock sound ranking                     |
| Extra RAM                        | Available through SGM, ADAM, and compatible clones depending on the tool/runtime        | Report separately; never count it as stock 1 KB RAM                                            |

This separation lets every solution show its extended-hardware strengths without weakening Amy Studio's deliberate original-hardware target.



## Development experience

| Area                       | Amy Studio                           | CVBasic                    | z88dk              | ugBASIC                    | devkitSMS         | PVCoLib                        |
|----------------------------|--------------------------------------|----------------------------|--------------------|----------------------------|-------------------|--------------------------------|
| Browser IDE                | Integrated                           | No official integrated IDE | No                 | IDE/web options advertised | No                | No                             |
| Source breakpoints         | Yes                                  | External emulator          | External debugger  | IDE-dependent/unverified   | External emulator | External emulator              |
| ASM stepping/rewind        | Integrated GearColeco workflow       | External                   | External           | External                   | External          | External                       |
| RAM/VRAM/symbol inspection | Integrated                           | External                   | External           | External                   | External          | External/map symbols           |
| Cycle profiling            | Integrated                           | External/manual            | External/manual    | Unverified                 | External/manual   | External/manual                |
| Project graphics editors   | Integrated/configurable              | Separate tools             | Separate tools     | Resource conversion/IDE    | Separate tools    | <b>gfx2col</b> /separate tools |
| Automated ROM tests        | Corpus, checkpoints, runtime harness | Not integrated             | Buildable manually | Not established            | Not established   | Not established                |

This is Amy Studio's decisive advantage. A fair comparison should still label command-line-only Amy scripts separately from features directly accessible in the IDE.

### Amy Studio leads in ColecoVision integration

Amy's strongest distinction is not one isolated keyword. It connects source editing, asset conversion, compression, assembly optimization, ROM execution, source breakpoints, rewind, memory and VRAM inspection, controller configuration, profiling, and automated tests in one ColecoVision-focused workflow. Its overlays, typed state machines, BCD, fixed-point support, runtime-checked wide integers, collision helpers, and BIOS-aware input selection are also substantial language-level strengths.

### CVBasic leads in portability and established BASIC simplicity

CVBasic has a compact QBasic-like surface, a mature ColecoVision backend, many complete game examples, spinner support, sprite flicker support, music commands, and optional ROM banking. It also targets many related machines. It does not offer Amy's records, overlays, wide numeric model, integrated debugger, or asset/debug pipeline.

Official source: [nanochess/CVBasic](https://github.com/nanochess/CVBasic)

### z88dk leads in general C and toolchain breadth

z88dk provides full C data structures, pointers, mature compilers, assemblers, linkers, libraries, compression utilities, and many Z80 targets. This flexibility also exposes more low-level choices to the programmer. Its generic graphics or library facilities must not be mistaken for an Amy-style ColecoVision game API without target-specific proof.

Official source: [z88dk/z88dk](https://github.com/z88dk/z88dk)

### ugBASIC has the most ambitious portable high-level vocabulary

ugBASIC documents nonblocking animation, movement, paths, image conversion, sprites, sound, and many modern BASIC constructs. This makes it especially relevant when considering future Amy animation ergonomics. However, its multi-target manual describes capabilities that can differ by backend. Generated ColecoVision Z80 and runtime behavior must be measured before declaring those features equivalent.

Official sources: [ugBASIC site](https://ugbasic.iwashere.eu/) and [spotlessmind1975/ugbasic](https://github.com/spotlessmind1975/ugbasic)

### devkitSMS offers a compact C game-library workflow

devkitSMS combines SDCC with small game-oriented libraries and asset tools. **SGLib\_CV** and **PSGLib\_CV** make it relevant to ColecoVision even though much of the surrounding documentation is SMS-oriented. Only APIs present in the CV headers and libraries should enter the final score.

The local **SGLib\_CV** source confirms that its ColecoVision NMI scans direction/fire mode and numpad/right-trigger mode, preserves current and previous states, and implements **SG\_getKeysStatus**, **SG\_getKeysPressed**, **SG\_getKeysHeld**, **SG\_getKeysReleased**, and **SG\_readCVNumPad**. These input capabilities are source-verified for ColecoVision, not inferred from SMSlib.

Official source: [sverx/devkitSMS](https://github.com/sverx/devkitSMS)

### PVColLib offers broad ColecoVision-focused C APIs

PVColLib is a ColecoVision-focused SDCC library and development kit. Its official repository includes the compiler toolchain, VDP and controller APIs, sprites, sound/music support, compression routines, MegaCart, SGM, Phoenix, and F18A facilities. The three stock fixtures here use its official build flags and linked library. Local headers and examples verify both controller ports and keypads, both spinners, 32 sprites, NMI sprite swapping, hitbox collisions, BIOS-style sound tables, sequenced music with concurrent SFX areas, direct VRAM operations, MegaCart, SGM, Phoenix, and F18A support. Its RLE bitmap path was validated byte-for-byte in VRAM; the available Pletter and DAN paths are not credited with a benchmark size until a compatible stream passes the same exact runtime test.

Official source: [alekmaul/pvcollib](https://github.com/alekmaul/pvcollib)

### NewColeco remains a strong historical baseline

The recovered SDCC branch uses Amy's original **CRTCv**, **CVLIB**, and GETPUT 1.1 libraries. Its prebuilt ASxxxx objects link successfully with the locally available SDCC 3.8 toolchain after using the modern **.rel** extension; no ABI or library source change was required. The three new fixtures complete 180 NTSC frames in GearColeco. Warrior is decompressed directly to VRAM with GETPUT's MDK-RLE routine and matches all 12,288 target table bytes and all 49,152 pixels.

This is not an unrelated seventh competitor: it is the documented historical ancestor of Amy Studio. Its 795-byte Hello, 3,643-byte DAN2 bitmap, and 932-byte controller monitor show that the old BIOS-aware and link-only-what-is-used philosophy was already effective. Amy Studio improves those results while adding the Amy language, integrated assets, diagnostics, and debugging.

## Next measurement suite

The comparison becomes stronger as equivalent game behaviors are implemented seven times. Current status and order:

1. **Controller snapshot: complete.** Six toolchains pass injected neutral, keypad, direction, fire, and release states; ugBASIC's VDP R7 update remains the documented exception.
2. **Sprite/metasprite stress: complete.** Seven toolchains build and boot the shared fixture. Amy's native metasprite path is runtime-equivalent, uses zero permanent RAM, preserves explicit layer priority, and is smaller than the manual renderer in all five profiles.
3. **Tile animation: next.** Update a small Graphics II region every frame without corruption and measure VRAM bytes per frame.
4. **State update.** Run an actor array, collision checks, timers, and state dispatch.
5. **Sound authoring: functional and still evolving.** BIOS commands and Tiny Sound sequences can be inspected, auditioned, edited, imported, and written back; UX and audio-parity QA continue.
6. **Compression: payload and cycle suites complete.** Fourteen codecs have exact VRAM proof; representative streams now include linked decoder bytes and GearColeco cycle measurements.
7. **Visible Hello: complete.** Keep it separate from the minimal runtime fixture so font/text costs remain explicit.

The first fixture is now in **competition/benchmarks/controller-input**. Initial build facts:

- Amy: Off 413, Safe 409, Balanced 408, Aggressive 408, and Experimental 406 ROM bytes. The fixture has five explicit result bytes plus Amy's selected runtime state. All controller capabilities are selected from actual source usage.
- z88dk +coleco -02: 2,117 bytes of useful binary before its cartridge image is padded to 32,768 bytes. Its CRT/BSS map reaches **\$717A**; the five result bytes occupy **\$701F-\$7023**.
- CVBasic, ugBASIC, and devkitSMS are now locally built and measured below.

Controller fixture ROM results

The build keeps all five ROMs under `build/competition/controller-input/` so they can be loaded manually in Amy Studio ROM TEST & DEBUG or another ColecoVision emulator.

| Tool                 | Occupied ROM | ROM file | Entry  | GearColeco          |
|----------------------|--------------|----------|--------|---------------------|
| Amy Studio Balanced  | 408          | 408      | \$8120 | 120-frame boot PASS |
| devkitSMS / SDCC 4.5 | 1,337        | 16,384   | \$8024 | 120-frame boot PASS |
| CVBasic 0.9.2        | 1,483        | 8,192    | \$853D | 120-frame boot PASS |
| z88dk +coleco -02    | 2,117        | 32,768   | \$802A | 120-frame boot PASS |
| ugBASIC 1.18         | 5,337        | 16,384   | \$82BF | 120-frame boot PASS |

**Occupied ROM** compares compiler and runtime output. **ROM file** is the downloadable file size and includes each toolchain's padding. CVBasic is measured from **ROM\_END**, z88dk from its unpadded linked binary, ugBASIC from its code and data binaries, and devkitSMS from its Intel HEX span. The measurement never guesses by trimming trailing **\$00** or **\$FF** bytes.

The later seven-solution controller suite adds controlled input injection and assertions. Six solutions pass equivalent semantics; ugBASIC boots but does not update VDP R7 in this fixture.

For every fixture record source lines, compiler version, build time, ROM bytes, permanent RAM, worst-frame cycles, and runtime result. A smaller ROM that fails visually or changes behavior is not an optimization win.

## Ranked Amy gap plan

| Rank | Work                                                  | Value  | Effort | Risk        | Decision gate                                                                                             |
|------|-------------------------------------------------------|--------|--------|-------------|-----------------------------------------------------------------------------------------------------------|
| 1    | Close sound-editor fidelity and UX gaps               | High   | Medium | Medium      | Emulator-faithful preview, visible Tiny envelopes, reliable inline edits, undo, and byte-exact write-back |
| 2    | Add per-asset compression cost guidance               | High   | Medium | Low         | Show payload + decoder bytes and measured/estimated cycles without claiming one universal winner          |
| 3    | Seven-tool tile-animation benchmark                   | High   | Medium | Low         | Equivalent Graphics II workload, VRAM bytes/frame, cycles, and corruption oracle                          |
| 4    | State-update benchmark                                | High   | Medium | Low         | Equivalent actor arrays, collisions, timers, dispatch, ROM, RAM, and worst-frame cycles                   |
| 5    | Small explicit animation service                      | High   | Large  | Medium-high | Add only after benchmark evidence; zero linked cost when unused and visible RAM/cycle budget              |
| 6    | Nested aggregate 2D fields and final operand symmetry | Medium | Medium | Medium      | Direct record/overlay 2D fields already pass; add nesting only for a real game need                       |

## Completed gaps and next concrete work

Completed and runtime-guarded:

- the seven-tool sprite/metaspprite fixture and shared GearColeco SAT oracle;
- native `data ... metasprite layers N` and `set metasprite ...`, with zero permanent RAM and

savings of **33**, **33**, **18**, **17**, and **17** bytes versus explicit rendering in the five profiles;

- stable sprite ranges and flicker with protected high-priority sprites;
- direct primitive 2D fields in records and overlay parts, verified in all five profiles;
- ZX1 direct-to-VRAM with exact four-picture GearColeco VRAM proof;
- Studio BIOS/Tiny Sound inspection, continuous playback, synchronized channel lanes, inline

command/envelope editing, undoable write-back, import, and Web MIDI capture.

Next concrete work is **sound-editor closure**, not another sequencer rewrite: verify preview timing against GearColeco, expose every Tiny envelope that affects playback, and keep timeline edits, undo, source write-back, table indices, and shared tails consistent. In parallel, surface the new per-asset compression size/cycle evidence in Studio. Then build the tile-animation and state-update fixtures before designing a general animation service.